$$
\begin{aligned}
\mathcal{S}_t &= (h_t, c_t, A_t, R_t, O_t, b_t, V_t, E_t, \Theta_t),\\
N_t &= |V_t|, \quad h_t, c_t \in \mathbb{R}^{N_t}, \quad R_t, O_t \in \mathbb{R}^{3\times N_t},\\
m_t &= \frac{\mathrm{Bid}_t + \mathrm{Ask}_t}{2}, \quad s_t = \mathrm{Ask}_t - \mathrm{Bid}_t \geq 0,\\
\delta_t &> 0, \quad \Delta t_t > 0, \quad \varepsilon > 0,\\
x_t &= \begin{bmatrix}(m_t - m_{t-1})/\delta_t & s_t/\delta_t & \chi_t\end{bmatrix}^\top,\\
\sigma^\star_{t,k} &= \max(\sigma_{t,k}, \varepsilon), \qquad z_{t,k} = \mathrm{clip}\left(\frac{x_{t,k} - \mu_{t,k}}{\sigma^\star_{t,k}}, -7, 7\right),\\
\lambda_{i,t} &= 1 - e^{-\Delta t_t/\tau_i}, \quad \tau_i > 0,\\
h^-_{i,t+1} &= h_{i,t} + \lambda_{i,t}\left[\tanh\left(p_i^\top z_t + \sum_{j:(j,i)\in E_t} w_{ij} h_{j,t}\right) - h_{i,t}\right],\\
\gamma_{i,t} &= 1 - e^{-\Delta t_t/\tau^c_i}, \quad \tau^c_i > 0,\\
c^-_{i,t+1} &= c_{i,t} + \gamma_{i,t}\left[\tanh(0.5 h^-_{i,t+1} + 0.2 c_{i,t}) - c_{i,t}\right],\\
\ell_t &= R_t h^-_{t+1} + O_t c^-_{t+1} + b_t \in \mathbb{R}^3,\\
M_t &= \max_{k\in\{1,2,3\}} \ell_{t,k}, \qquad p_{t,j} = \frac{e^{\ell_{t,j} - M_t}}{\sum_{k=1}^3 e^{\ell_{t,k} - M_t}},\\
a_t &= \min \arg\max_{j\in\{1,2,3\}} p_{t,j},\\
\mathrm{dir}(a_t) &= \begin{cases} 1 & a_t = 1,\\ -1 & a_t = 2,\\ 0 & a_t = 3. \end{cases}
\end{aligned}
$$

$$
\begin{aligned}
\widehat{z}_t &= G_t(h^{-}_{t+1}), \quad v_t = \|z_t - \widehat{z}_t\|_2, \\
\overline{v}_t &= (1-\alpha)\overline{v}_{t-1} + \alpha v_t, \quad 0 < \alpha \le 1, \\
g_t &= \mathbf{1}\{v_t > \theta \ \wedge \ \overline{v}_t > \overline{\theta} \ \wedge \ N_t < N_{\max}\}, \\
B_t &= \begin{cases} \{u_t\}, & g_t = 1,\ u_t \notin V_t, \\ \varnothing, & g_t = 0, \end{cases} \\
p_{u_t} &= \frac{z_t}{\|z_t\|_2 + \varepsilon}, \quad h_{u_t,t+1} = c_{u_t,t+1} = 0, \\
\mathcal{N}^{\mathrm{in}}_{u_t} &= \varnothing, \quad \mathcal{N}^{\mathrm{out}}_{u_t} = \varnothing, \\
R_{t+1,:,u_t} &= r_{u_t}, \quad O_{t+1,:,u_t} = o_{u_t}, \quad \|r_{u_t}\|_2 + \|o_{u_t}\|_2 > 0, \\
\tau_{u_t} &> 0, \quad \tau^{c}_{u_t} > 0, \quad U_{u_t,t+1} = 0, \quad \mathrm{age}_{u_t,t+1} = 0, \\
y_{t,j} &\ge 0, \quad \sum_{j=1}^{3} y_{t,j} = 1, \quad 0 < \varepsilon_p < \tfrac{1}{3}, \\
\mathcal{L}_t &= -\sum_{j=1}^{3} y_{t,j} \log\big(\max(p_{t,j}, \varepsilon_p)\big), \\
\ell_t^{(-i)} &= \ell_t - R_{t,:,i} h^{-}_{i,t+1} - O_{t,:,i} c^{-}_{i,t+1}, \\
p_{t,j}^{(-i)} &= \frac{\exp(\ell_{t,j}^{(-i)} - M_t^{(-i)})}{\sum_{k=1}^{3} \exp(\ell_{t,k}^{(-i)} - M_t^{(-i)})}, \\
M_t^{(-i)} &= \max_k \ell_{t,k}^{(-i)}, \quad \mathcal{L}_t^{(-i)} = -\sum_{j=1}^{3} y_{t,j} \log\big(\max(p_{t,j}^{(-i)}, \varepsilon_p)\big), \\
q_{i,t} &= \mathcal{L}_t^{(-i)} - \mathcal{L}_t, \\
U_{i,t+1} &= (1-\beta) U_{i,t} + \beta q_{i,t}, \quad 0 < \beta \le 1.
\end{aligned}
$$

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\begin{aligned}
\widetilde{D}_t &= \{i \in V_t : U_{i,t+1} < \epsilon_i \ \wedge \ \mathrm{age}_{i,t} \geq T_{\min}\},\\
D_t &\subseteq \widetilde{D}_t, \qquad |D_t| \leq \max\{0, N_t + |B_t| - N_{\min}\},\\
1 &\leq N_{\min} \leq N_t \leq N_{\max},\\
V_{t+1} &= (V_t \setminus D_t) \cup B_t,\\
N_{t+1} &= N_t - |D_t| + |B_t|, \quad N_{\min} \leq N_{t+1} \leq N_{\max},\\
E(D_t) &= \{(i,j) \in E_t : i \in D_t \ \vee \ j \in D_t\},\\
E(B_t) &= \varnothing, \qquad E_{t+1} = (E_t \setminus E(D_t)) \cup E(B_t),\\
\Theta_t &= \{P_t, W_t, R_t, O_t, b_t, \tau_t, \tau_t^c, G_t\},\\
\Theta_t^\star &= \Theta_t - \eta_t \nabla_{\Theta_t} \mathcal{L}_t, \quad \eta_t > 0,\\
\Theta_{t+1} &= \mathrm{resize}(\Theta_t^\star, V_{t+1}, E_{t+1}),\\
R_{t+1}, O_{t+1} &\in \mathbb{R}^{3 \times N_{t+1}},\\
(h_{t+1}, c_{t+1}) &= \mathrm{resize}\left((h_{t+1}^-, c_{t+1}^-), V_{t+1}\right),\\
\mathrm{age}_{i,t+1} &= \mathrm{age}_{i,t} + 1 \quad (i \in V_t \setminus D_t),\\
\mu_{t+1,k} &= (1-\zeta)\mu_{t,k} + \zeta x_{t,k}, \quad 0 < \zeta \leq 1,\\
\nu_{t+1,k} &= (1-\zeta)\nu_{t,k} + \zeta (x_{t,k} - \mu_{t,k})^2,\\
\sigma_{t+1,k} &= \sqrt{\max(\nu_{t+1,k}, \varepsilon^2)}.
\end{aligned}
$$

$$
\begin{aligned}
\delta_t > 0, \quad s_t \geq 0, \quad d_t^{\min} \geq 0, \\
b_t^{\mathrm{buf}} &= \max(2\delta_t, 2s_t), \\
d_t &= \max(20\delta_t, 4s_t, d_t^{\min} + b_t^{\mathrm{buf}}), \\
n_t^{\mathrm{SL}} &= \left\lceil \frac{d_t}{\delta_t} \right\rceil, \quad \rho_t = n_t^{\mathrm{SL}} \delta_t, \\
\mathrm{SL}_t &= \begin{cases} P_e - \rho_t, & a_t = 1, \\ P_e + \rho_t, & a_t = 2, \\ \varnothing, & a_t = 3, \end{cases} \\
\kappa > 0, \quad n_t^{\mathrm{TP}} &= \left\lceil \frac{\kappa \rho_t}{\delta_t} \right\rceil, \\
\mathrm{TP}_t &= \begin{cases} P_e + n_t^{\mathrm{TP}} \delta_t, & a_t = 1, \\ P_e - n_t^{\mathrm{TP}} \delta_t, & a_t = 2, \\ \varnothing, & a_t = 3, \end{cases} \\
V_t^{\mathrm{tick}} > 0, \quad Q_t^{\mathrm{step}} > 0, \quad Q_t^{\min} \geq 0, \quad Q_t^{\max} \geq Q_t^{\min}, \\
R_t^{\max} > 0, \quad C_t^{\mathrm{risk}} \geq 0, \\
Q_t^{\mathrm{raw}} &= \frac{\max(0, R_t^{\max} - C_t^{\mathrm{risk}})}{n_t^{\mathrm{SL}} V_t^{\mathrm{tick}}}, \\
Q_t^{\mathrm{floor}} &= Q_t^{\mathrm{step}} \left\lfloor \frac{\min(Q_t^{\mathrm{raw}}, Q_t^{\max})}{Q_t^{\mathrm{step}}} \right\rfloor, \\
Q_t^{\mathrm{pos}} &= \begin{cases} Q_t^{\mathrm{floor}}, & a_t \neq 3,\ Q_t^{\mathrm{floor}} \geq Q_t^{\min},\ Q_t^{\mathrm{floor}} > 0, \\ 0, & \text{otherwise}, \end{cases} \\
\mathrm{Risk}_t &= n_t^{\mathrm{SL}} V_t^{\mathrm{tick}} Q_t^{\mathrm{pos}} + \mathbf{1}\{Q_t^{\mathrm{pos}} > 0\} C_t^{\mathrm{risk}} \leq R_t^{\max}.
\end{aligned}
$$

$$
\begin{aligned}
C_t &= C_t^{\text{commission}} + C_t^{\text{slippage}} + C_t^{\text{other}} \geq 0, \\
f_t &= \frac{C_t}{V_t^{\text{tick}} Q_t^{\text{pos}}}, \qquad Q_t^{\text{pos}} > 0, \\
r_t &= \begin{cases} \dfrac{P_x - P_e}{\delta_t} - f_t, & a_t = 1,\ Q_t^{\text{pos}} > 0, \\ \dfrac{P_e - P_x}{\delta_t} - f_t, & a_t = 2,\ Q_t^{\text{pos}} > 0, \\ 0, & \text{otherwise}, \end{cases} \\
\Pi_t &= \begin{cases} r_t V_t^{\text{tick}} Q_t^{\text{pos}}, & Q_t^{\text{pos}} > 0, \\ 0, & Q_t^{\text{pos}} = 0, \end{cases} \\
\Pi_t &= \text{dir}(a_t) \frac{P_x - P_e}{\delta_t} V_t^{\text{tick}} Q_t^{\text{pos}} - C_t, \quad Q_t^{\text{pos}} > 0, \\
\mathcal{F}_t &= \mathcal{F}_{\text{learn}} \circ \mathcal{F}_{\text{broker}} \circ \mathcal{F}_{\text{order}} \circ \mathcal{F}_{\text{decision}} \circ \mathcal{F}_{\text{neuron}}.
\end{aligned}
$$